

# Okta Account Security

A sign-in investigation view for repeated authentication failures followed by a successful login.

Status: Splunk dashboard demonstration documented

## Executive summary

Configured an Okta Account Security dashboard panel titled “Repeated Failed Sign-Ins Followed by Success.” The synthetic scenario groups authentication activity by user and client IP, then surfaces multiple failures followed by a success for analyst triage. The view is designed to prompt validation of user, device, MFA, and session context.

## Scope and method

Fictional banking identity-monitoring scenario with synthetic Okta-style sign-in events in Splunk. No real Okta tenant or customer identity was accessed.

## Results

| Signal                   | Why it matters                                                                                                    |
|--------------------------|-------------------------------------------------------------------------------------------------------------------|
| Repeated failed sign-ins | May indicate password guessing, mistyped credentials, stale automation, or user error.                            |
| Success after failures   | Raises priority for review, especially from a new IP, device, or location; does not alone prove account takeover. |
| Dashboard panel          | Groups events by user and client IP and presents attempt / failure counts for analyst review.                     |

# Evidence screenshots

Figure 1. Synthetic Okta-style sign-in dataset

```
akachi@akachi-ThinkPad-T440s:~$ mkdir -p ~/okta-account-security
cat > ~/okta-account-security/okta_signins.csv <<'EOF'
published,user,client_ip,country,outcome,event_type
2026-10-04T09:00:00Z,alex.chen@northstarbank.example,198.51.100.10,CA,SUCCESS,user.session.start
2026-10-04T09:05:00Z,alex.chen@northstarbank.example,198.51.100.10,CA,SUCCESS,user.authentication.verify
2026-10-04T10:10:00Z,jordan.lee@northstarbank.example,198.51.100.22,CA,SUCCESS,user.session.start
2026-10-04T11:20:00Z,sam.patel@northstarbank.example,198.51.100.31,CA,SUCCESS,user.session.start
2026-10-04T12:00:00Z,alex.chen@northstarbank.example,203.0.113.77,NG,FAILURE,user.authentication.verify
2026-10-04T12:01:00Z,alex.chen@northstarbank.example,203.0.113.77,NG,FAILURE,user.authentication.verify
2026-10-04T12:02:00Z,alex.chen@northstarbank.example,203.0.113.77,NG,FAILURE,user.authentication.verify
2026-10-04T12:03:00Z,alex.chen@northstarbank.example,203.0.113.77,NG,FAILURE,user.authentication.verify
2026-10-04T12:04:00Z,alex.chen@northstarbank.example,203.0.113.77,NG,SUCCESS,user.session.start
2026-10-04T12:05:00Z,alex.chen@northstarbank.example,203.0.113.77,NG,SUCCESS,user.authentication.verify
2026-10-04T13:10:00Z,jordan.lee@northstarbank.example,198.51.100.22,CA,SUCCESS,unc -l ~/okta-account-security/okta_signins.csvxample,198.51.100.31,CA,FAILURE,us
13 /home/akachi/okta-account-security/okta_signins.csv
```

okta\_signins.csv created with fictional northstarbank.example users and documentation-range IPs (198.51.100.0/24, 203.0.113.0/24). The sequence for alex.chen from 203.0.113.77 contains four failures followed by a success. Terminal capture, 18:08.

Figure 2. Detection search in Splunk SPL

The screenshot shows the Splunk Enterprise search interface. The search bar contains the following SPL query:

```
source="okta_signins.csv"
| stats count as attempts
  count(eval(outcome="FAILURE")) as failures
  count(eval(outcome="SUCCESS")) as successes
  values(country) as countries
  by user client_ip
| where failures >= 4 AND successes >= 1
```

The search results are displayed in a table with the following columns: user, client\_ip, attempts, failures, successes, and countries. The results show one entry for alex.chen@northstarbank.example with 6 attempts, 4 failures, 2 successes, and country NG.

| user                            | client_ip    | attempts | failures | successes | countries |
|---------------------------------|--------------|----------|----------|-----------|-----------|
| alex.chen@northstarbank.example | 203.0.113.77 | 6        | 4        | 2         | NG        |

The search groups sign-ins by user and client\_ip and keeps combinations with at least four failures and one success. One row returned: 6 attempts, 4 failures, 2 successes, country NG. Splunk capture, 18:12.

Figure 3. Okta Account Security dashboard panel

The screenshot shows the Splunk Enterprise dashboard panel titled "Okta Account Security". The panel displays a table with the same data as Figure 2, showing repeated failed sign-ins followed by success for alex.chen@northstarbank.example.

| user                            | client_ip    | attempts | failures | successes | countries |
|---------------------------------|--------------|----------|----------|-----------|-----------|
| alex.chen@northstarbank.example | 203.0.113.77 | 6        | 4        | 2         | NG        |

Saved dashboard panel "Repeated Failed Sign-Ins Followed by Success" rendering the same result for analyst triage. The panel surfaces a lead; it does not revoke sessions or confirm compromise. Splunk capture, 18:23.

# Okta Account Security | Analyst notes

## Recommended follow-up

1. Validate the user, source, device, location, MFA outcome, and session events against authorized identity logs.
2. Ask the identity owner to verify the activity using the approved contact path.
3. If confirmed suspicious, follow the organization's access-response policy for session revocation and credential or MFA remediation.
4. Tune thresholds using normal sign-in patterns, known VPN egress, service accounts, and travel context.

## Limitations

Synthetic demonstration data only. The dashboard does not revoke sessions, reset credentials, or prove compromise. IP geolocation is approximate and should not be treated as identity evidence. No live Okta deployment or production telemetry is represented.

## Evidence basis

A Splunk screenshot showed the dashboard title and repeated-failure panel. The event-level evidence was synthetic; this report does not claim a real compromise.

## Skills demonstrated

Identity telemetry, Splunk SPL, authentication triage, dashboard design, and safe response recommendations.

Portfolio integrity note: This report describes an educational demonstration. It is not a client engagement, production security assessment, or certification of control effectiveness.